Spending Fewer Atoms: Slack-Free One-Hot Clustering of the Capacitated Vehicle Routing Problem on an Analog Neutral-Atom Register

Team Qualies — DuQuantum Hackathon, Pasqal challenge (Track A primary, Track B supplementary)

25 October 2026 · repository `qhack/` (one-command reproduction: `python scripts/make.py demo`)

Abstract. We put the clustering half of the capacitated vehicle routing problem (CVRP) onto Pasqal's `AnalogDevice` with *no slack atoms at all*. Customers are assigned to vehicles with one-hot variables; the capacity inequality is handled by unbalanced penalization, which on one-hot strings collapses exactly to a load-balance coupling. Every coupling is positive and every linear term is equal, so the model runs on a single global detuning with NK atoms: 8, 30 and 60 on tiers S, M and L, against 14, 45 and 80 for the textbook slack encoding. The report is organised around the chain of improvements that took this model from paper to a working register. A structure-aware embedder—rigid 5 µm polygons fitted on the one-hot subspace—raised the probability that a raw shot is the model-optimal clustering from 3–13 % to 62–90 %, and a pulse scan raised it further to **89–99 % on all five tier-S instances**, with feasibility 0.95–1.00. On larger registers we derived and enforced a physical bound on the final detuning, which took tier M from 0 % to 98–100 % decodable shots, and matched the register geometry and drive to the $K = 4$ energy scale, which took tier L from under 1 % to **14–23 % capacity-feasible shots**. Every gap is computed on feasible decodes only, and everything that did not work is listed.

1 What we built

- **A slack-free model** (Sec. 4): one-hot assignment plus unbalanced penalization; the *one-hot identity* removes every non-uniform linear term exactly. 25 % of the atom budget saved on L.
- **A structure-aware embedder** (Sec. 6): it asks the register to reproduce only what the model needs on valid strings, not every pairwise coupling.
- **Two derived physical bounds** for one-hot cliques on a planar register, $\delta_f < U_{\min}$ and the $1/8$ coupling collapse of a $K = 4$ square, each turned into a code rule.
- **A reproducible pipeline**: seeded, certified instances; exact references; committed registers; Pulser-validated sequences; four emulators cross-checked; 79 result rows; plus two verified routing encodings for Track B.

2 Problem and decomposition (Derive 2.1)

Customers $i \in C = \{0, \dots, N-1\}$ have locations $p_i \in [0,100]^2$ and integer demands d_i. There are K identical vehicles of capacity Q and a depot fixed at the centre $(50,50)$ before clustering; travel costs c_{uv} are Euclidean. A valid plan puts every customer on exactly one vehicle (an **equality**), loads no vehicle beyond Q (an **inequality**), and orders each vehicle's visits; we minimise total distance, and an unused vehicle costs nothing. Encoding vehicle *and* visit order needs $\sim N^2K$ binaries (32/300/900), so we use cluster-first/route-second: the register decides which customers share a vehicle, and each cluster is routed by exact Held–Karp. The price is the *decomposition gap*, which we report on every instance.

3 What the register minimises (Derive 3.1)

At the end of the pulse $\Omega = 0$, so the device Hamiltonian is diagonal and assigns each bitstring x the cost

$$E_{\text{dev}}(x) = -\delta_f \sum_i x_i + \sum_{i<j} \frac{C_6}{r_{ij}^6} x_i x_j. \tag{1}$$

This is a QUBO with three built-in rules. The **laser** sets one linear coefficient, $-\delta_f$, shared by every atom ($|\delta| \le 125.66$ rad/µs). The **positions** set every coupling, all positive, at most $C_6/(5\,\mu\text{m})^6 = 55.4$ rad/µs and never exactly zero. C_6 is fixed physics. So any model we run must have uniform linear terms and positive couplings.

Table 1: Variable count (Derive 4.1), generated by `scripts/variable_count.py`; $B = \lceil \log_2(Q+1) \rceil$ slack bits per vehicle.

Encoding	S	M	L	fits 80?
Assignment only (NK)	8	30	60	yes
+ binary slack ($NK + KB$)	14	45	80	yes (L = cap)
Ours, `onehot-ub` (NK)	**8**	**30**	**60**	**yes**

4 The model (A1): one-hot, no slack

Variables. $x_{ik} = 1$ iff customer i rides on vehicle k; NK binaries and nothing else. **Objective.** Total pairwise distance inside each cluster, $f(x) = \sum_k \sum_{i<j} d_{ij} x_{ik} x_{jk}$: quadratic, positive, and additive over clusters, so an exact set-partition dynamic programme certifies the model optimum on every tier. **Exactly-one.** The penalty $A \sum_i (1 - \sum_k x_{ik})^2$ expands to a linear term $-A$ on every bit and a coupling $+2A$ between the K bits of each customer. **Capacity.** With slack $g_k = Q - L_k \ge 0$, $L_k = \sum_i d_i x_{ik}$, unbalanced penalization [1] adds $\sum_k [-\lambda_1 g_k + \lambda_2 g_k^2]$ and needs no auxiliary bits. Its linear part is $\sum_i c_i \sum_k x_{ik}$ with $c_i = d_i(\lambda_1 - 2\lambda_2 Q + \lambda_2 d_i)$. On every one-hot string $\sum_k x_{ik} = 1$, so this part equals the constant $\sum_i c_i$. We call this *the one-hot identity*: it lets us drop every non-uniform linear term, and λ_1 with it, without changing the model where it lives. What remains is

$$2\lambda_2 \sum_k \sum_{i<j} d_i d_j x_{ik} x_{jk} = \lambda_2 \sum_k L_k^2 + \text{const}, \tag{2}$$

a soft load-balance term. The final cost is

$$E(x) = \sum_k \sum_{i<j} w_{ij} x_{ik} x_{jk} + A \sum_i \Big(1 - \sum_k x_{ik}\Big)^2, \quad w_{ij} = d_{ij} + 2\lambda_2 d_i d_j, \tag{3}$$

with all couplings positive and all linear terms $-A$, so $\delta_f = \alpha A$ realises it with no DMM. **Weights.** $A = \max_i \sum_{j \ne i} w_{ij} + 1$ provably makes every global minimiser one-hot. $\lambda_2 = \kappa R/(2Q)$, $R = \max_i \sum_j d_{ij}$, $\kappa = 0.15$, chosen by brute force on 40 toy seeds and verified on all five shipped S instances. Because capacity is soft, *we score only feasible decodes* and always report the feasibility rate.

Cost (Table 1). We save 6/15/20 atoms and never read out a bit we throw away. The real cost is geometric: a customer's K atoms must sit 5 µm apart, so every other atom is almost equidistant from them, and couplings meant to be zero come out as large as the wanted ones. Section 6 is our answer.

5 Instances (A2) and references

A seeded generator draws a balanced assignment *first* and then demands that fit each vehicle's remaining capacity, so every instance is feasible by construction. That assignment ships as the certificate; `cvrp.validator` regenerates each instance from its seed and re-checks it. We use S seeds 0–4 and M, L seeds 0–2, with load ratios ρ = 0.64–0.86 / 0.70–0.95 / 0.60–0.71 and $d_{\max}/Q$ = 0.43–0.57 / 0.40 / 0.30–0.33. **References** follow brief Sec. 9.3. CPLEX could not be installed, so C^* is the best of `qubo-solver`'s simulated annealing on our QUBO and a constrained annealer on the CVRP (10 seeded restarts each, 60 s limit, parameters in each instance file). On tier S an exhaustive search agrees 5/5, so those are *proven optima*; M and L are *best known*.

6 Embedding (A3): ask the register for less

Where we started. A generic least-squares placement of the coupling graph gave a relative coupling error $\|U/\alpha - J\|_F / \|J\|_F$ of 0.95, and on S_seed04 the register's own ground state was an infeasible string.

The idea. The model only lives on one-hot strings, where each pair of customers contributes exactly one coupling. So the cross-vehicle

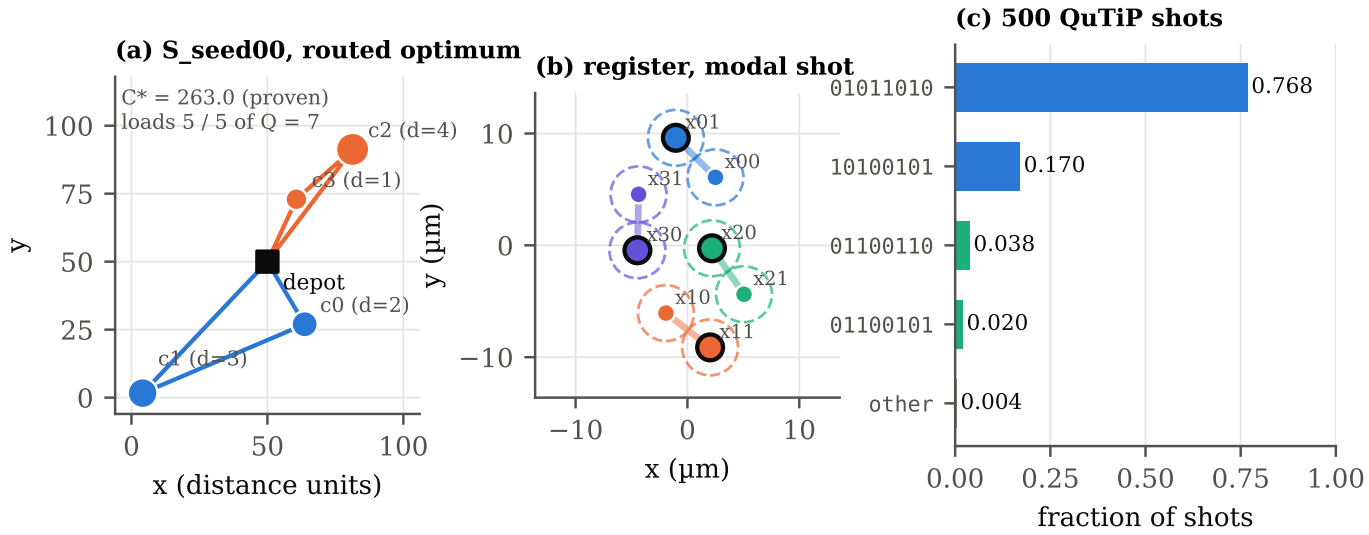


Figure 1: The pipeline on the headline instance. (a) S_seed00 with its proven-optimal clustering, routed exactly ($C^* = 263.0$). (b) The 8-atom register: four 5 µm dimers, one per customer; bold atoms are excited in the most frequent shot `01011010`, which decodes to that clustering. (c) 500 QuTiP shots: the two labelings of the optimal clustering (blue) make up 93.8 % of shots; 99.8 % are capacity-feasible.

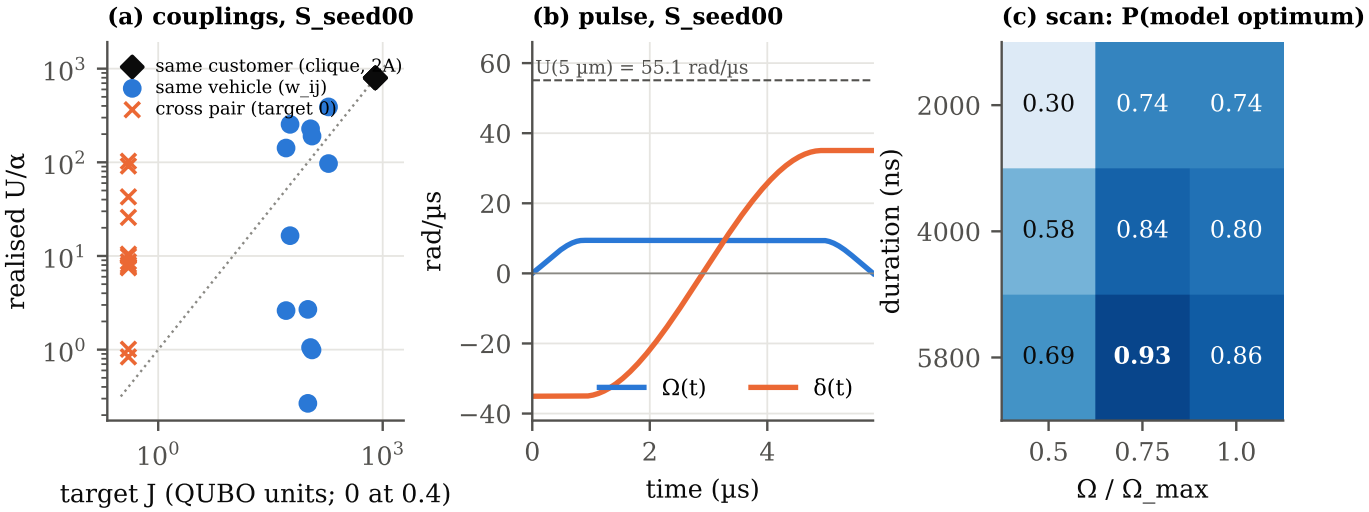


Figure 2: Embedding and pulse on S_seed00. (a) Target vs realised couplings by pair class: cliques are exact, same-vehicle couplings approximate, and cross pairs (target 0) are large but nearly constant on valid strings, which is all the model needs. (b) The committed pulse: three-point Ω ramp at $0.75\,\Omega_{\max}$ and a linear δ sweep to the fitted +35.1 rad/µs over 5800 ns, below the 55.1 rad/µs clique bound. (c) The duration × amplitude scan (mean over S_seed00/01/04).

couplings do not have to vanish; they only have to be constant. Only the *effective* coupling $U(i,k;j,k) - U(i,k;j,l \neq k)$ must equal αw_{ij}.

The embedder (`cvrp/embed_onehot.py`) makes each customer a rigid regular K-gon of side 5 µm, which realises the exactly-one clique exactly. It then fits only the polygon centres and orientations so that the device energies of all one-hot strings match αE_{QUBO}, keeps every single-flip defect above its parent, and fits δ_f as a free parameter. Registers are cached and committed so every row is reproducible. On tier S the full-matrix error falls to 0.21–0.32, and the register's own ground state *is* the model optimum on 5/5 instances (Fig. 2a).

Two bounds we derived and enforced. In an isolated K-gon, one excitation beats two only if $0 < \delta_f < U_{\min}$, where $U_{\min}$ is the *weakest* coupling inside the polygon. For $K \leq 3$ that is 55.1 rad/µs, and tier S fits land at 35–38. On tier M the fit ran to the device maximum, 125.66 rad/µs, and *no* shot was decodable; enforcing the bound fixed it (Table 2). For a $K = 4$ square the diagonal is $\sqrt{2}$ longer, and $1/r^6$ turns that into a factor 8: $U_{\min} = 6.9$ rad/µs. We caught this before reporting any L run and now use the weakest pair of the actual polygon.

7 Pulse (A4)

We use the baseline of brief Sec. 8.2: a three-point Ω ramp (15 % up, 70 % plateau, 15 % down) and a linear δ sweep during the plateau from far below resonance to the fitted δ_f. Every sequence is built with Pulser on `AnalogDevice`, serialised, and validated against the device object. We scanned duration and amplitude on three S instances and scored each setting by the probability of reaching the optimum, the best gap and feasibility, never the mean cost (Fig. 2c). Longer was always better, and $0.75\,\Omega_{\max}$ beat full power, which disturbs the 55 rad/µs cliques more. We committed $T = 5800$ ns and $\Omega = 0.75\,\Omega_{\max}$; rescaling δ_f by 0.8 or 1.25 was worse. Two instances that were not in the scan (S_seed02/03) reach the optimum in 99 % and 96 % of shots. Emulators: QuTiP for every tier-S headline row, cross-checked by emu-sv and an in-house exact state-vector integrator; emu-mps for M and L, with bond dimension χ and peak χ recorded on every row.

8 Results (A4, brief Table 3)

Table 2 lists every improvement we made, each measured before and after on the same instances. Table 3 gives the best final row per tier with the metrics the brief asks for; all 79 rows are in `results.csv`, one per (instance, method, seed), with atoms, feasibility, best model and CVRP gaps on feasible decodes, P(reach C^*), P(gap $\leq \varepsilon$) for $\varepsilon \in \{0, 1, 5, 10\}$ % (we quote 5 %), fidelity, reference status, χ, DMM (never used) and repair label.

Tier S. On all five instances the register returns the model-optimal clustering in **0.94/0.95/0.99/0.96/0.89** of raw shots, with feasibility 0.95–1.00 and a 0 % model gap (Figs. 1, 3). The CVRP gap is 0 % on S_seed00/01/02. On S_seed03/04 it is 15.6/21.7 % while the model gap stays at 0 %: the register found the closest clustering, which is not the shortest-route one. That loss is the decomposition gap, and it is a property of the objective, not of the sampler. For context

Table 2: The improvement ladder. Each step was measured on the same instances and committed registers before and after the change (raw shots, 500 per row, no repair). P(opt) = fraction of shots that are the model-optimal clustering. Values are per instance in the order listed.

#	What we changed	Why	Measured on	Before → after
1	Textbook slack → one-hot + unbalanced penalization	Slack bits cost atoms and give non-uniform linear terms that a global detuning cannot realise	all tiers	Atoms 14/45/80 → **8/30/60**; linear terms non-uniform → uniform
2	Generic embedder → polygons fitted on one-hot subspace	"Zero" cross couplings came out as strong as the wanted ones	S_seed00/01/04	Coupling error 0.95 → 0.21–0.32; P(opt) 0.13/0.07/0.03 → **0.90/0.88/0.62**; feasibility 0.22/0.09/0.06 → 0.99/0.88/0.99
3	Pulse 4000 ns at $\Omega_{\max}$ → 5800 ns at $0.75\,\Omega_{\max}$	Longer sweep; gentler drive on the clique couplings	S_seed00/01/04	P(opt) 0.90/0.88/0.62 → **0.94/0.95/0.89**; feasibility 0.99/0.88/0.99 → 1.00/0.95/1.00
4	Enforce $\delta_f < U_{\text{clique}}$	Fit ran to 125.66 rad/µs, rewarding two excitations per triangle	M_seed00/02	Decodable 0/0 → **0.99/1.00**; feasible 0/0 → 0.044/0.446
5	Bound from the weakest polygon pair	A square's diagonal coupling is 8× weaker than its side	tier L	δ_f 35.8 → 4.5 rad/µs; the mis-set run was discarded before reporting
6	5 µm → 8 µm spacing between polygons	Caps spurious couplings at 3.3 rad/µs; defect gaps 1.1 → 2.4 rad/µs	L_seed00/01/02	Optimum's one-hot rank on L_seed00: 16 534 → 1 795
7	Drive matched to the $K = 4$ energy scale, Ω 0.75 → $0.15\,\Omega_{\max}$	The whole one-hot landscape spans a few rad/µs	L_seed00/01/02	Feasible 0.008/0.004/0 → **0.208/0.140/0.226**; best CVRP gap 80/75/— → **47/40/36 %**
8	Bond dimension $\chi = 32 \to 64$	Convergence of the 30-atom emulation	M_seed01	Best CVRP gap 9.2 → **5.9 %**; peak χ 14 (converged)

Table 3: Best final row per tier (brief Table 3 metrics). Raw shots, no repair, no DMM; every gap on feasible decodes only; $\varepsilon = 5\,\%$. Fidelity = $\|U/\alpha - J\|_F/\|J\|_F$ on the full matrix (one-hot-subspace value in parentheses).

instance	method / backend	atoms	feas.	model gap	CVRP gap	P(C^*)	P(gap≤5 %)	fidelity	reference	emulation
S_seed00	polygons + tuned pulse	8	1.00	0.0 %	0.0 %	0.94	0.94	0.26 (0.38)	proven	QuTiP, exact
S_seed02	polygons + tuned pulse	8	1.00	0.0 %	0.0 %	0.99	0.99	0.31 (0.28)	proven	QuTiP, exact
M_seed01	polygons, bounded δ_f	30	0.39	33.8 %	5.9 %	0.00	0.00	0.08 (0.96)	best known	emu-mps χ=64
L_seed02	8 µm polygons, $0.15\,\Omega_{\max}$	60	0.23	47.1 %	36.3 %	0.00	0.00	0.51 (1.05)	best known	emu-mps χ=32

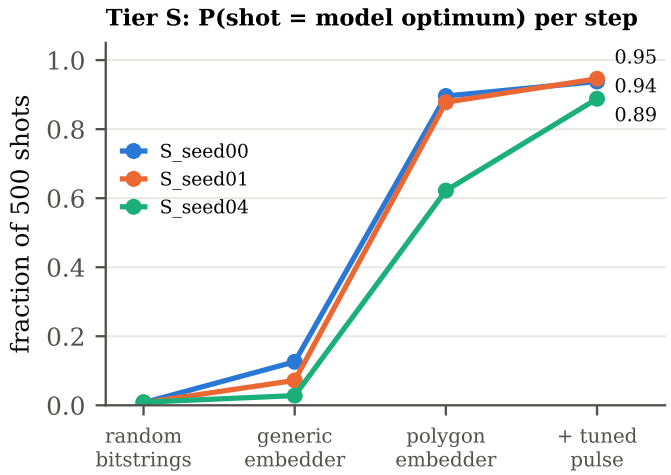


Figure 3: Tier S, step by step (QuTiP, 500 shots per point; Table 2 rows 2–3). The structure-aware embedder is the largest single gain; the pulse scan closes most of the remaining gap, most of all on S_seed04.

(brief Sec. 9.2), random bitstrings reach the optimum in about 1 % of shots, uniformly random valid assignments in 12.5 %, and Pasqal's `qubo-solver` pipeline in 0–79 %. *One ablation shaped the design:* the tighter provable penalty $A = R_w/K$ keeps the optimum as the ground state but drops P(opt) from 0.94 to 0.03 on S_seed00, because it shrinks the energy margin the sweep needs. We therefore kept the larger A.

Tier M (30 atoms). After the δ_f bound (ladder step 4), 98–100 % of shots are one-hot and 4.4/38.8/44.6 % are capacity-feasible. The best CVRP gaps are 29.9/5.9/16.0 % (M_seed01 at $\chi = 64$). Enumerating all 3^{10} one-hot strings shows where the remaining loss sits: the capacity term costs nothing on two instances, and the pulse is not the bottleneck, because the sampled best shots beat the register's own best feasible one-hot string. The loss is the register's ordering of clusterings (the optimum sits at rank 4 192–12 966 of 59 049), the geometric limit of $K = 3$ triangles.

Tier L (60 atoms, $K = 4$). The weak square diagonal caps δ_f at 4.5 rad/µs and squeezes the whole one-hot landscape into a few rad/µs. With the 5 µm packing and the tier-S drive, only 0.8/0.4/0 % of shots were feasible. Spacing the polygons to 8 µm and matching Ω to the new energy scale (ladder steps 6–7, Fig. 4) raised feasibility to **20.8/14.0/22.6 %** and brought the best CVRP gaps to **46.9/40.4/36.3 %**, with 37–41 % of shots one-hot. The amplitude is a sharp optimum: at $0.10\,\Omega_{\max}$ no shot decodes, and above 0.25 performance falls off again. Emulation: emu-mps, $\chi = 32$, dt=10 ns, precision 10^{-5}, peak χ 17–32.

Follow-up studies. Decomposition gaps are 0/0/0/15.6/21.7 % (S), 0/0/6.9 % (M) and 17.6/8.2/13.1 % (L). An exact comparison of three clustering objectives shows that a depot-aware one, $\text{MST}(S) + 2\min_{i\in S} d_{0i}$, cuts the L decomposition gaps to 0/0/7.5 % but worsens every M instance; it is the natural next objective for L. On eight deliberately capacity-binding S instances (four held out, selection frozen first), a stronger balance weight $\kappa = 0.6$ raises mean feasibility from 0.48 to 0.80 but lowers mean P(reach C^*) from 0.39 to 0.19, so it stays an ablation. A bounded classical repair makes 98–100 % of M shots feasible; it is reported separately and never mixed into raw metrics.

9 Track B: routing (B1–B3)

B1. For a cluster of n customers plus the depot, $y_{i,t} = 1$ if customer i is visited at position t. The objective is $\sum_t \sum_{i\neq j} c_{ij} y_{i,t} y_{j,t+1}$ plus the two depot legs; each customer takes one position and each position one customer (both equalities). **B2, two encodings.** (i) *Depot-fixed position*, n^2 bits. The depot gets no variables; its legs are folded into the first and last transition weights, $w(i,j,t) = c_{ij} + [t{=}1]c_{0i} + [t{=}n-1]c_{j0}$. The energy then equals the tour length exactly on every valid string, and every linear term is $-2A$. This saves $2n+1$ bits over the textbook $(n+1)^2$. (ii) *Undirected edge*, $n(n+1)/2$ bits: one bit per edge, a degree-2 penalty, and each edge's cost split between its two endpoints so the linear terms stay uniform ($-6A$). It is exact for $n \leq 4$; for $n \geq 5$ two disjoint triangles can be cheaper than any tour, which we demonstrate explicitly. Both encodings were verified by exhaustive enumeration on 18 cases (Fig. 5a). **B3.** Clusters with $n \leq 3$ are trivial and routed exactly; $n = 4$ clusters run both encodings (10 and 16 atoms, exact state-vector emulation, 1000 shots each); $n \geq 5$ are routed classically and labelled. On the tier-M clusterings, 12 clusters were routed: 7 trivial, 5 quantum attempts, one raw valid quantum tour (20.4 % routing gap) and 4 labelled classical fallbacks, so 1 of 5 non-trivial clusters was routed quantumly. End-to-end gaps are 19.6 % (M_seed01) and 16.0 % (M_seed02), split exactly into decomposition, clustering and routing loss (Fig. 5b). Both routing QUBOs are correct; the limit is the generic embedding, and a structure-aware routing register, built the same way as Track A's, is the next step.

10 What did not work

- **Generic embedding of the coupling graph:** error 0.95, P(opt) 3–13 %, and an infeasible ground state on S_seed04. Fixed for

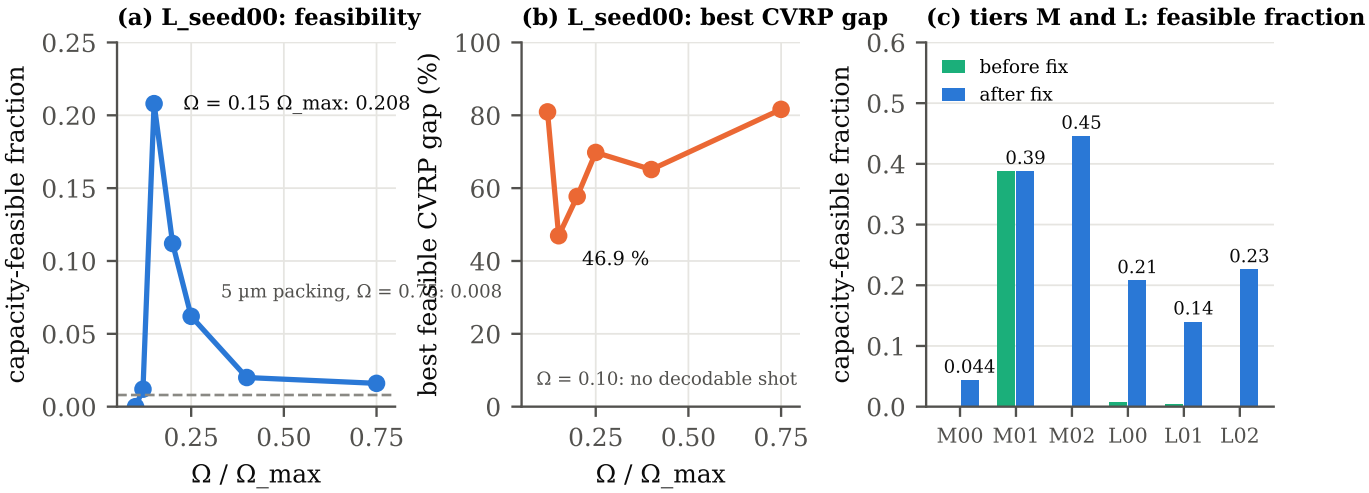


Figure 4: Tiers M and L (emu-mps $\chi = 32$, 500 shots). (a,b) L_seed00 at 8 µm polygon spacing: matching the drive to the compressed $K = 4$ energy scale ($\Omega = 0.15\,\Omega_{\max} = 1.9$ rad/µs against 2.4 rad/µs defect gaps) is a sharp optimum. It moves L_seed00 from 0.8 % feasible with an 80 % best CVRP gap (5 µm packing, tier-S drive) to 20.8 % and 47 %. (c) Capacity-feasible fraction before and after the fixes (M: ladder step 4; L: steps 6–7).

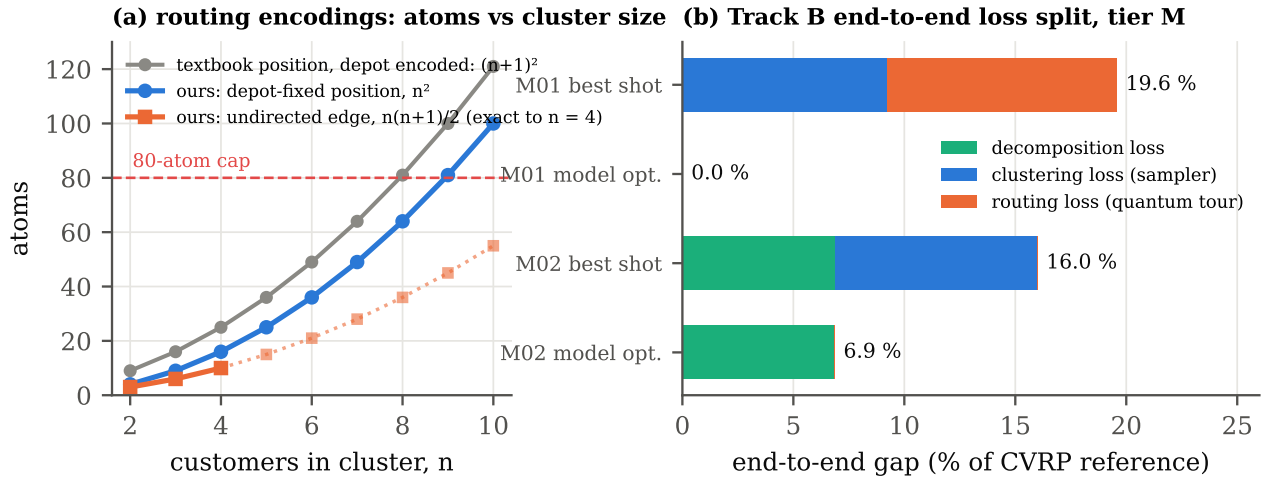


Figure 5: Track B. (a) Atoms needed by each routing encoding as the cluster grows; the edge model is exact up to $n = 4$. (b) End-to-end loss split into decomposition + clustering + routing, an identity verified to 10^{-9} on every row.

$K = 2$ by the polygon fit; for $K \geq 3$ the ordering of clusterings is still distorted.

- **Unbounded δ_f on tier M:** 0 of 500 shots decodable on two instances. The first L run, set with the triangle bound, would have favoured two opposite corners and was discarded.
- **No single κ works on all toy seeds** (best 35/40): where capacity binds, "closest" and "most balanced" conflict, and a soft term cannot force $L_k \leq Q$.
- **The weight scan showed no trend**, because the embedding optimiser lands in a different local minimum each time. It is also not bit-identical across platforms (51 % vs 90 % from one seed), so every register is committed.
- **The tight A** is provably sound but removes the device's margin ($0.94 \to 0.03$).
- **qubo-solver's annealer** returned the same non-optimal string on three S instances, and its uncapped emu-mps run on M ($\chi > 300$) was stopped after 80 min.
- **Track B:** one valid quantum tour in 10 000 shots with the generic embedding.
- **Plumbing:** no CPLEX wheel; a cp1252 console crash on "δ"; a | in a file name lost a record on Windows; a stale PDF was briefly reported as fresh; a sync reset a parallel edit. All are fixed and logged in `docs/DID_NOT_WORK.md`.

11 Conclusions and next steps

We showed that the capacity inequality can go onto an analog register with *zero* slack atoms, all couplings positive and one global detuning, at NK atoms on every tier. Each step from paper to device was measured: the polygon embedder and the pulse scan took tier S to 89–99 % optimal shots; a derived detuning bound made tier M decodable; and a register and drive matched to the $K = 4$ energy scale took tier L from under 1 % to 14–23 % feasible shots. The next improvements are in the register: an encoding of "which vehicle" that avoids K-gons (for example hierarchical $K = 2$ bisection, which keeps every tier-S property), a depot-aware objective for L, and a structure-aware routing register for Track B.

Use of AI. Claude and Codex were used throughout: typing up and, at the team's request, drafting derivations, and writing code, scans, figures, tables and prose. A fresh-context AI reviewer audited physics, algebra and fairness. Every number here is generated from run records; algebra is checked by exhaustive tier-S enumeration and device legality by Pulser. The team chose the model, the protocol and the claims, and owns all of them (`docs/AI_USE.md`).

References

[1] J. A. Montañez-Barrera, D. Willsch, A. Maldonado-Romo, K. Michielsen, *Unbalanced penalization: a new approach to encode inequality constraints of combinatorial problems for quantum optimization algorithms*, Quantum Sci. Technol. 9, 025022 (2024), arXiv:2211.13914.